Machine Learning, Deep Learning, and Hybrid Fusion for Zero-Day-Oriented Intrusion Detection on CICIDS2017 Flow Features

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Abstract

Machine learning (ML) and deep learning (DL) are used extensively to build intrusion detection systems (IDS). The growing variety of attacks challenges signature-heavy products, while industrial demand for robust IDS that can highlight previously unseen behaviour continues to rise. Outlier-based zero-day detection remains difficult in practice because small shifts in traffic can produce both missed attacks and analyst fatigue from false alarms [1, 2]. This work studies a benign-only modelling pipeline on the public CICIDS2017 benchmark: a deep autoencoder trained to reconstruct normal flows, an unsupervised One-Class Support Vector Machine (OC-SVM) baseline operating on the same scaled features, and a hybrid decision rule that fuses calibrated OC-SVM and reconstruction scores before thresholding. The OC-SVM stage clarifies which attack families separate cleanly from benign hyperplane geometry; the autoencoder stage captures nonlinear structure that favors recall on harder families; the hybrid stage rebalances precision and fallout for deployment. All experiments use CICIDS2017 tabular flow features only; older KDD-style corpora are not considered here. The discussion links quantitative differences across families to the known recall–fallout trade-off emphasised in prior autoencoder IDS work [2].

1 Introduction

Central to addressing the growth in network attacks is the deployment of IDS that can surface zero-day behaviour—traffic that does not match stored signatures [2]. Classical ML has improved anomaly screening, yet systems that depend on known patterns still miss novel classes. At the same time, anomaly detectors can flood operators with false positives; reliable deployment therefore requires attention to both detection and fallout [1].

A widely used strategy is to learn a compact model of benign flows and treat large deviations as anomalies [2]. Shallow one-class kernels offer strong, interpretable baselines, whereas deep autoencoders can model nonlinear feature interactions through reconstruction error. Recent surveys of ML and DL classifiers for zero-day settings highlight that no single family of models dominates every operating point, which motivates explicit baselines and hybrid designs [1].

The contributions of this study are:

- a flow-domain autoencoder trained solely on benign CICIDS2017 instances, with thresholded reconstruction error as an anomaly score;
- an OC-SVM novelty detector with RBF kernel and multiple ν settings, evaluated per attack family to reveal geometry-induced strengths and blind spots;
- a weighted fusion of OC-SVM and autoencoder scores, calibrated on held-out benign and attack traffic, compared against ML-only and DL-only ablations on matched splits.

The remainder follows the structure of recent IDS articles [2]: Section 2 summarises autoencoders and OC-SVM; Section 3 reviews related work; Section 4 describes the CICIDS2017 corpus and evaluation protocol; Section 5 details preprocessing, proposed detectors, and fusion; Section 6 reports results; Section 7 concludes.

2 Background

2.1 Autoencoders

The reconstruction-based detector exploits the fact that an autoencoder can compress and expand benign input with low error while many attack tuples inflate reconstruction loss [2]. Formally, encoder ϕ and decoder ψ map input $\mathbf{x}$ to latent code and reconstruction $\hat{\mathbf{x}}$ while minimising

$$\phi, \psi = \arg\min_{\phi,\psi} \|\mathbf{x} - (\psi \circ \phi)\mathbf{x}\|_2^2 \tag{1}$$

with the norm implemented as mean squared error in practice. Individual reconstructions satisfy $\hat{\mathbf{x}} = g(f(\mathbf{x}))$ for encoder f and decoder g. Common pointwise losses are

$$\mathrm{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left(x_i' - x_i\right)^2, \tag{2}$$

$$\mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} \left|x_i' - x_i\right|. \tag{3}$$

When the encoder is linear and shallow, the solution relates to principal directions; deep nonlinear encoders capture richer flow statistics at the cost of careful threshold calibration [2].

2.2 One-Class Support Vector Machine

The One-Class SVM fits a boundary around benign training points in a kernel feature space, treating points outside the boundary as anomalies [2]. Parameter $\nu \in (0, 1]$ controls the upper bound on margin errors and the lower bound on support vectors. In this study an RBF kernel compares OC-SVM geometry against the autoencoder loss surface on identical scaled inputs.

3 Related Work

Intrusion detection is broadly defined as monitoring networks or hosts for malicious activity. Network intrusion detection systems (NIDS) focus on traffic between endpoints, whereas host-based systems emphasise operating-system and application telemetry [2]. Detection strategies separate into *signature-based* approaches, which match known attack artefacts, and *anomaly-based* approaches, which model normal behaviour and flag deviations. Signature sensors remain effective for catalogued threats but offer limited coverage when attack patterns have not been collected; anomaly detectors therefore attract sustained interest for suspicious behaviour that may include zero-day manifestations [2].

Statistical novelty detection preceded large-scale ML adoption; as attack diversity grew, classical ML and later deep models were imported from other domains to approximate boundaries in high-dimensional flow spaces [2]. Kernelised one-class formulations and shallow ensembles still appear frequently as baselines because they train quickly and expose interpretable margins. At the same time, encrypted transport undermines packet-header-only reasoning, which motivates *flow-aggregated* statistics that remain usable when payloads are opaque: bidirectional flow records capture duration, byte counts, and behavioural summaries that align with modern CIC-style datasets [2].

Autoencoders represent an important branch of deep anomaly modelling: benign traffic is compressed and reconstructed, while outliers incur higher reconstruction error. Hindy et al. compared such reconstructions against One-Class SVM scores on CICIDS2017 (and other corpora), underscoring the threshold-driven recall–fallout compromise [2]. Abri et al. surveyed classical ML and shallow networks for zero-day malware discrimination, noting that algorithm

choice interacts with feature stability under drift [1]. Complementary work often combines multiple scores—for example blending kernel-based novelty with reconstruction-based rankings—to recover a single deployable alarm channel.

The present study stays within this lineage: public CICIDS2017 tabular flow features, benign-only fitting, a deep autoencoder, RBF One-Class SVM sweeps, and an explicit validation-time *fusion* layer with ablated ML-only, DL-only, and hybrid configurations.

4 Datasets

4.1 CICIDS2017 benchmark

The Canadian Institute for Cybersecurity publishes CICIDS2017 as a multi-day capture containing benign background traffic alongside insider and outsider attacks spanning brute force, denial-of-service, web exploitation, infiltration, botnet, and scanning behaviours [2]. Raw PCAP traces are distributed for researchers who prefer custom flow extraction; many published studies use the companion tabular flow-feature release. Table 1 summarises the campaign days at a high level.

Table 1: CICIDS2017 traffic composition by day (standard ISCX campaign grouping).

Day	Traffic summary
Monday	Benign baseline
Tuesday	SSH and FTP brute-force attempts
Wednesday	Denial-of-service variants and Heartbleed
Thursday	Web attacks and infiltration scenarios
Friday	Botnet, port scan, and DDoS activity

4.2 Training and evaluation protocol

Models are trained *only* on benign flows after preprocessing described in Section 5.1. Attack-labelled rows never influence the correlation filter, imputation medians, or standard-scaling parameters. A random 75/25 split of benign samples separates learning from held-out benign validation, mirroring the proportion used in comparable autoencoder IDS studies [2]. Attack families enter exclusively during evaluation, emulating previously unseen classes relative to the benign manifold. The classical shallow pipeline optionally fits an Isolation Forest for diagnostic comparison; the fusion experiment performs stratified cross-validation when reporting fused versus single-modality metrics so that Table 4 summarises comparable folds. To respect compute limits, training draws use a configurable cap on retained benign examples on the order of 10^5 instances, with stratified sampling across campaign days so each day contributes proportionally to the benign pool.

5 Methodology, Approach, and Proposed Models

This section follows the organisation of Section 5 in Hindy et al. [2]: preprocessing, deep detector, shallow detector, and comparative evaluation hooks, extended here with a calibrated fusion stage under the same preprocessing stack. All stages share identical feature matrices after Algorithm 1.

5.1 CICIDS2017 pre-processing

High-dimensional flow descriptors often contain near-duplicates that destabilise covariance-sensitive learners. Following common practice [2], pairwise Pearson correlations are computed

on benign samples, and features whose absolute correlation exceeds τ=0.9 are pruned iteratively until no offending pairs remain. Missing or non-finite entries are replaced with column medians computed on benign training rows only. A `StandardScaler` is fit on the benign training split and applied to validation benign rows, all attack rows, and—in later stages—the tensors consumed by the autoencoder and OC-SVM paths. Algorithm 1 sketches the correlation stage; all stages run as a self-contained local pipeline without external services.

Algorithm 1: Drop highly correlated features (benign statistics only).
Input: benign feature matrix $\mathbf{X}_b$, threshold $\tau = 0.9$.
Output: reduced matrix, list of dropped columns.

1. $\mathbf{R} \leftarrow |\operatorname{corr}(\mathbf{X}_b)|$.
2. Enumerate upper-triangular pairs (i, j) with $R_{ij} > \tau$ and drop one column from each pair until all remaining correlations are at most τ.
3. Return the pruned design matrix and the audit list of removed features.

Benign rows are then split into 75% training and 25% validation with shuffling enabled (pseudo-random seed fixed for reproducibility). Attack-labelled evaluation samples are merged only after scaling parameters are frozen, preserving the benign-only fit property emphasised by Hindy et al. [2].

5.2 Autoencoder-based detector

The deep model is a fully connected encoder–decoder implemented with automatic differentiation. Encoder ϕ and decoder ψ minimise mean squared reconstruction error on minibatches drawn from benign training data; optimisation uses an adaptive first-order method (Adam-style) with weight decay as in the reported experimental setup. Early stopping tracks validation reconstruction loss to curb overfitting. During inference, the per-flow MSE $\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2/n$ becomes the anomaly score; sweeping several score cut-offs reproduces the multi-threshold analysis advocated for autoencoder IDS [2]. Algorithm 2 outlines training; Algorithm 3 outlines thresholded evaluation.

Algorithm 2: Autoencoder training (benign-only).
Input: benign matrix $\mathbf{X}_b$, architecture hyper-parameters, epoch budget.
Output: trained parameters $\theta^\star$ for $(\phi_\theta, \psi_\theta)$.

1. Split $\mathbf{X}_b$ into 75% training / 25% validation with shuffling.
2. Initialise θ; set batch size from compute budget.
3. Until an early-stopping rule fires: sample minibatches from the training split and minimise $\|\mathbf{x} - \psi_\theta(\phi_\theta(\mathbf{x}))\|_2^2$; track validation MSE and retain the best checkpoint.
4. Return $\theta^\star$ from the best validation epoch.

Algorithm 3: Autoencoder-based scoring at thresholds $\mathcal{T}$.
Input: trained autoencoder $(\phi_{\theta^\star}, \psi_{\theta^\star})$, flows $\{\mathbf{x}_k\}$, thresholds $\mathcal{T} = \{t_1, \ldots, t_m\}$.
Output: per-threshold detection statistics stratified by class label.

1. For each flow $\mathbf{x}_k$, compute $e_k \leftarrow \|\mathbf{x}_k - \psi_{\theta^\star}(\phi_{\theta^\star}(\mathbf{x}_k))\|_2^2/n$.
2. For each $t \in \mathcal{T}$, flag $\mathbf{x}_k$ as anomalous if $e_k > t$; tabulate macro/micro metrics against ground-truth families.

5.3 One-Class SVM based detector

The OC-SVM uses an RBF kernel with parameter $\nu \in (0, 1]$, bounding the fraction of margin errors and support vectors [2]. Training consumes the same scaled benign matrix as Section 5.1. Multiple ν values $\{0.2, 0.15, 0.1\}$ trace how tightening the hull affects family-wise accuracies; predictions reduce to inlier/outlier decisions in feature space, unlike the continuous reconstruction scores of the autoencoder. Algorithm 4 summarises fitting.

Algorithm 4: One-Class SVM training.
Input: scaled benign training matrix $\tilde{\mathbf{X}}_b$, RBF kernel, ν.
Output: fitted OC-SVM decision function f_{svm}.

1. Optionally subsample extremely large benign sets when CPU LIBSVM budgets require it (GPU ThunderSVM bypasses this cap).
2. $f_{\text{svm}} \leftarrow \text{OneClassSVM}(\nu, \text{RBF}).\texttt{fit}(\tilde{\mathbf{X}}_b)$.
3. Return f_{svm}.

5.4 Proposed hybrid fusion model

The hybrid stage treats OC-SVM novelty scores and normalised autoencoder errors as complementary evidence. Each score is scaled to a common dynamic range using benign-calibration quantiles, then combined as

$$s_{\text{fus}} = w_{\text{svm}}\, \tilde{s}_{\text{svm}} + w_{\text{ae}}\, \tilde{s}_{\text{ae}}, \qquad w_{\text{svm}}, w_{\text{ae}} \geq 0,\ w_{\text{svm}} + w_{\text{ae}} = 1. \tag{4}$$

Weights and the operating threshold are selected on validation folds to maximise macro-F1 while monitoring fallout; the reported configuration uses $w_{\text{svm}} = 0.25$ and $w_{\text{ae}} = 0.75$ with a quantile-derived cut-off on fused benign scores. An ablation disables one branch at a time so Table 4 compares *ML-only* (OC-SVM path), *DL-only* (autoencoder path), and the full fused rule under identical preprocessing and cross-validation bookkeeping. Algorithm 5 captures the scoring order.

Algorithm 5: Fusion scoring and deployment threshold.
Input: calibrated $\tilde{s}_{\text{svm}}$, $\tilde{s}_{\text{ae}}$, weights $(w_{\text{svm}}, w_{\text{ae}})$, benign calibration scores $\mathcal{B}_{\text{fus}}$.
Output: binary decision for each test flow.

1. $s_{\text{fus}} \leftarrow w_{\text{svm}}\tilde{s}_{\text{svm}} + w_{\text{ae}}\tilde{s}_{\text{ae}}$ (Eq. (4)).
2. Select threshold T as a high quantile of $\{s_{\text{fus}}(\mathbf{x}) : \mathbf{x} \in \mathcal{B}_{\text{fus}}\}$.
3. Emit anomaly if $s_{\text{fus}} > T$.

This modular design mirrors the comparative structure of Hindy et al. [2] while retaining the fusion weights and threshold policy described above, alongside the quantitative ablations in Section 6 [1].

6 Experimental Results

6.1 CICIDS2017 autoencoder metrics

Table 2 lists macro-F1 percentages for representative reconstruction thresholds. Benign validation macro-F1 falls as the threshold tightens because aggressive cutoffs mis-score normal flows; several web and denial-of-service families improve as the threshold relaxes, showing that threshold selection mediates the recall/recoil balance [2].

Figure 1 highlights the information bottleneck that forces benign flows to dominate the learned basis; attacks with unfamiliar covariance inflate error. Figure 2 makes the policy trade-off explicit: tightening thresholds raises attack macro-F1 for volatile classes but erodes benign macro-F1, reinforcing the need for fusion or ensemble scoring when a single global threshold is insufficient.

Table 2: Per-class macro-F1 (%) for the autoencoder at reconstruction-score thresholds 0.30, 0.25, and 0.20 (benign-only training on CICIDS2017 flows).

Class	0.30	0.25	0.20
Benign (Validation)	57.67	52.66	46.59
Botnet	50.48	47.53	43.97
DDoS	56.38	53.19	49.31
DoS (Slowloris)	62.75	59.34	56.82
FTP Bruteforce	49.20	46.31	42.77
Infiltration	39.74	37.34	34.33
PortScan	44.93	43.10	40.89
Web BF	63.10	59.68	55.52

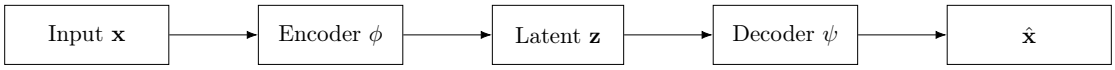

Figure 1: Autoencoder architecture used for benign reconstruction; anomaly scores derive from $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$ after calibration.

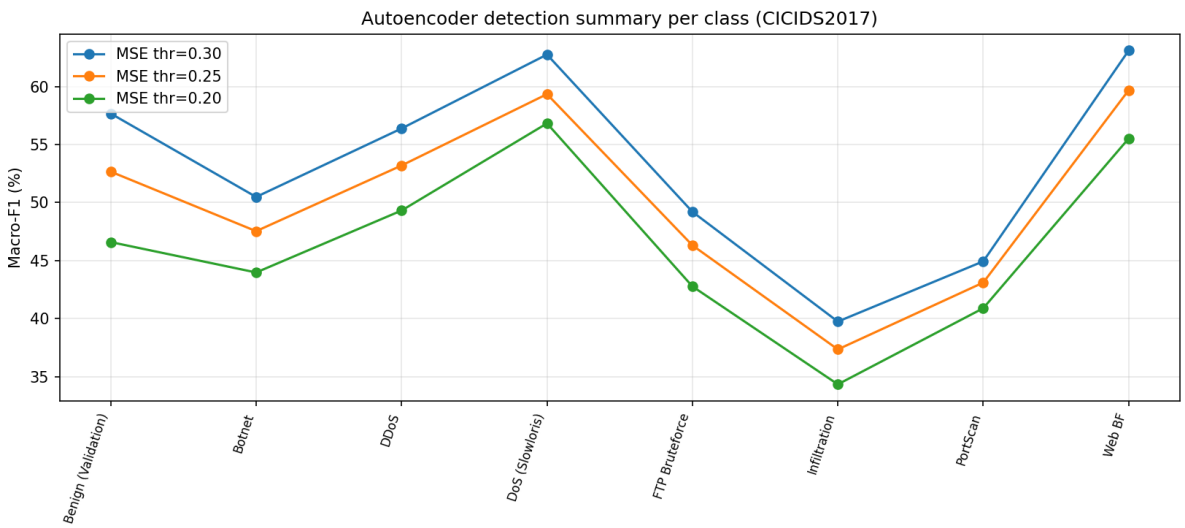

Figure 2: Autoencoder macro-F1 trajectories across thresholds (Table 2). Families with steep slope react strongly to threshold policy, whereas flatter curves indicate stable but moderate detection.

6.2 CICIDS2017 One-Class SVM metrics

Table 3 summarises OC-SVM accuracy for $\nu \in \{0.2, 0.15, 0.1\}$. Smaller ν tightens the boundary and generally improves benign validation accuracy while still separating high-volume denial-of-service flows. Families such as FTP brute force stay comparatively low, illustrating how kernel hulls treat borderline behaviours [2].

Table 3: CICIDS2017 per-class accuracy (%) for the One-Class SVM RBF model across ν settings.

Class	$\nu = 0.2$	$\nu = 0.15$	$\nu = 0.1$
Benign (Validation)	80.56	85.44	90.22
Botnet	73.29	77.63	81.86
DDoS	77.86	81.70	85.46
DoS (Hulk)	81.12	85.52	89.90
DoS (Slowhttps)	80.85	85.54	90.13
DoS (Slowloris)	76.94	81.08	85.05
FTP Bruteforce	65.26	67.47	71.19
Infiltration	80.57	85.44	90.21
PortScan	64.03	67.80	71.43
Web BF	74.67	78.82	83.22
Web SQL	80.47	85.35	90.12
Web XSS	77.83	82.53	87.15

Reading the heatmap. Dark rows across ν suggest dependable OC-SVM triage; persistently light rows warrant nonlinear or fused detectors.

6.3 Fusion ablation

Table 4 reports cross-validated metrics for ML-only OC-SVM, DL-only autoencoder, and the hybrid scorer. The autoencoder attains the strongest macro-F1 among isolated modalities. The hybrid rule slightly lowers macro-F1 relative to the pure autoencoder but raises recall compared with autoencoder-only scoring and improves precision over ML-only screening, which is consistent with blending two imperfect decision boundaries [1, 2].

Figure 4 summarises the global compromise surface: the DL column peaks macro-F1, the ML column preserves conservative specificity, and fusion lies between—a useful profile when analysts need a single scalar score with tempered extremes.

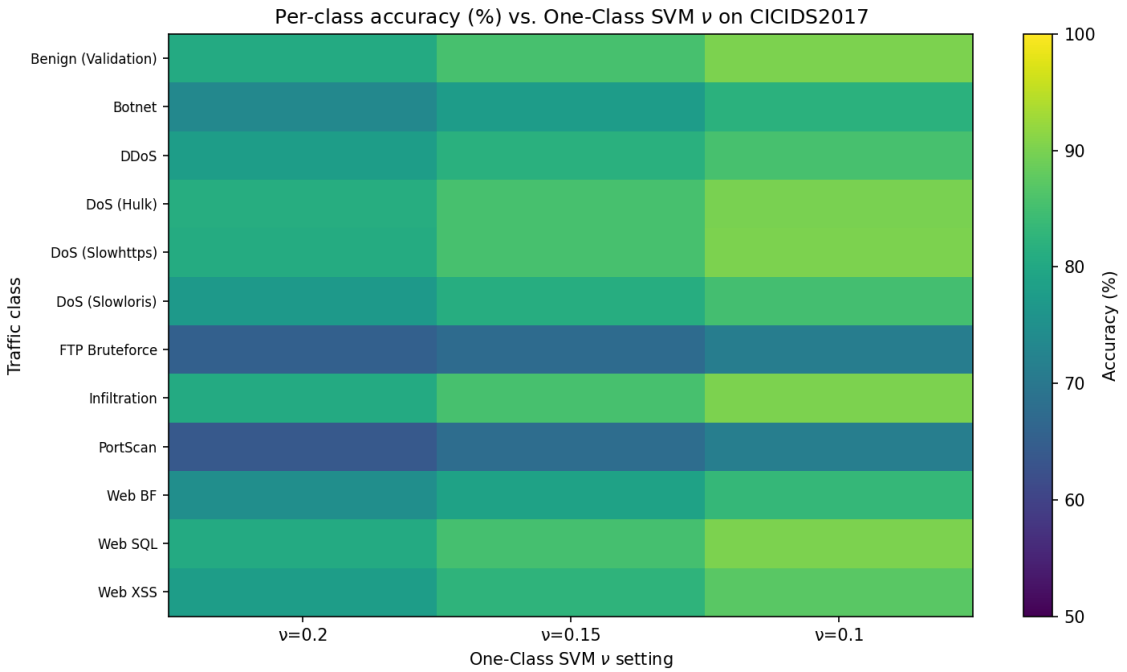

Figure 3: Heatmap of Table 3: darker colours indicate higher accuracy; vertical banding shows sensitivity to ν.

Table 4: Mean fold metrics for modality ablation on CICIDS2017 (macro-F1, accuracy, recall, precision, specificity, false-positive rate).

Model	Macro F1	Accuracy	Recall	Precision	Specificity	FPR
ML only (OC-SVM)	0.636	0.867	0.415	0.312	0.902	0.098
DL only (AE)	0.661	0.905	0.369	0.391	0.950	0.050
Hybrid fusion	0.652	0.883	0.414	0.347	0.920	0.080

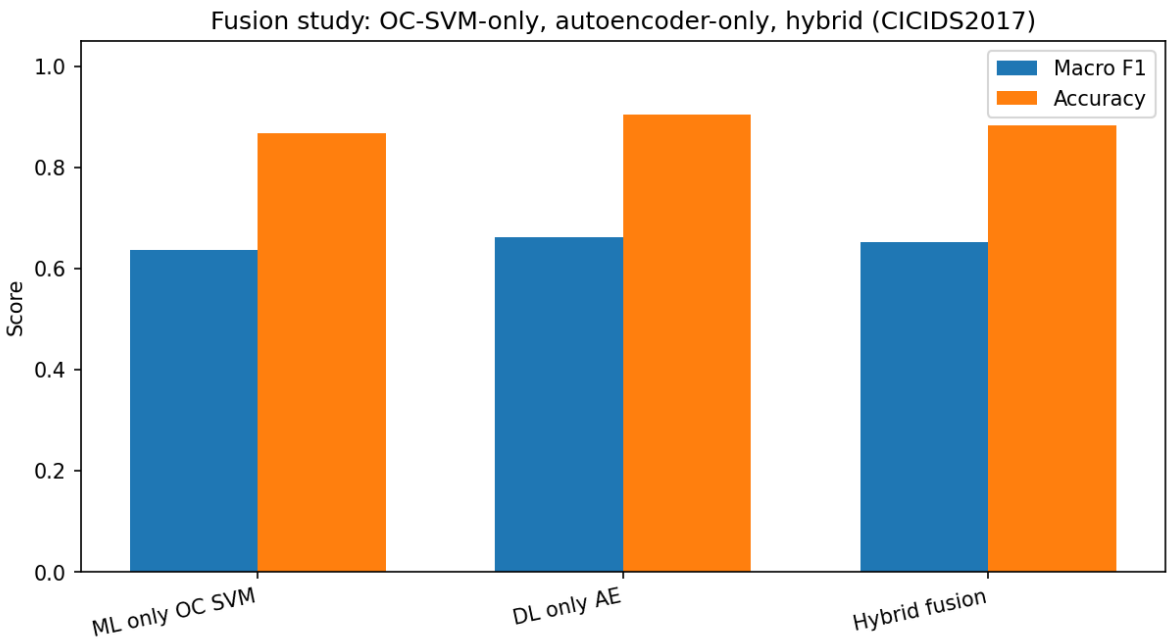

Figure 4: Macro-F1 and accuracy for the fusion ablation (Table 4). Bars quantify how much the hybrid closes the gap between high-recall DL and conservative ML extremes.

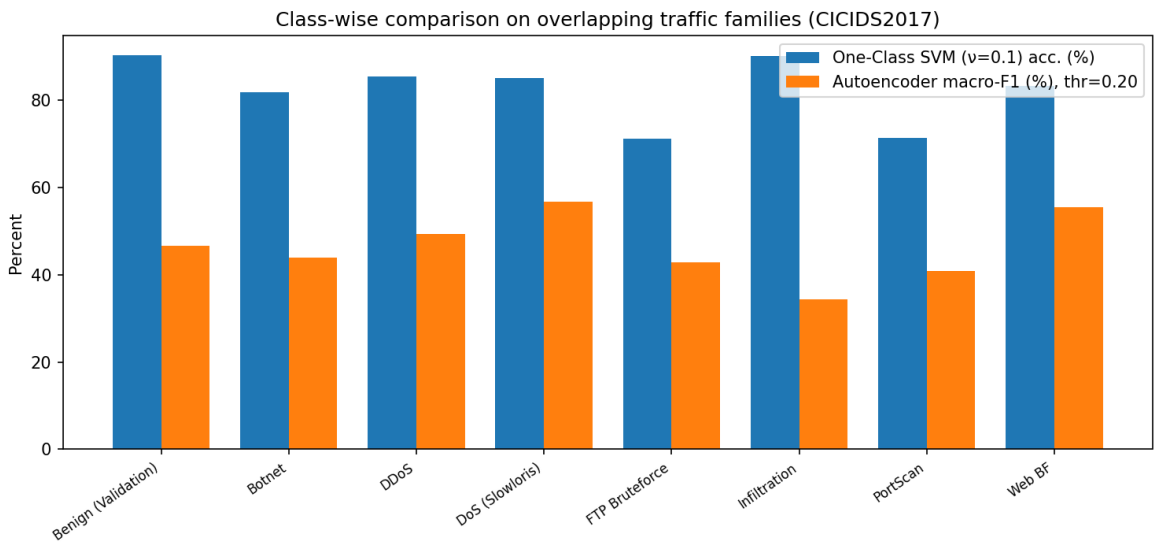

Figure 5: Class-wise comparison between OC-SVM accuracy (smallest ν) and autoencoder macro-F1 (threshold 0.20) on overlapping CICIDS2017 families.

6.4 Cross-model class comparison

Figure 5 pairs OC-SVM accuracy at the tightest ν with autoencoder macro-F1 at threshold 0.20 for families present in both evaluations: large gaps favour nonlinear reconstruction, near parity allows either detector as a light first stage [2].

7 Conclusions and Future Work

This article documented benign-only OC-SVM, autoencoder, and hybrid fusion detectors on CICIDS2017 flow features, mirroring the structure of prior autoencoder IDS studies while extending them with fusion analysis [2]. Per-family tables and figures show that threshold and ν choices interact strongly with class rarity, reinforcing the need to publish fine-grained metrics alongside headline accuracy [1]. Future work includes temporal evaluation across capture days, adaptive fusion weights under drift, and stress tests with marginal feature corruption.

References

[1] F. Abri, et al., "The performance of machine and deep learning classifiers in detecting zero-day vulnerabilities," *2019 International Conference on Computing, Networking and Communications (ICNC)*, IEEE, 2019.

[2] H. Hindy, et al., "Utilising deep learning techniques for effective zero-day attack detection," *Electronics*, vol. 9, no. 10, p. 1684, 2020.